Benedetta **Perrone**

PhD Researcher · KU Leuven

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Experience

KU Leuven – ESAT/STADIUS Division — *Leuven, Belgium*

PhD Researcher – EEG-based Brain-Computer Interfaces — *Nov. 2025 – Present*

- Developing calibration-free EEG-based brain-computer interfaces for decoding attention to naturalistic auditory stimuli.
- Funded by an FWO PhD Fellowship (Strategic Basic Research) from Nov. 2026.

Politecnico di Torino – Department of Automation and Informatics — *Turin, Italy*

Postgraduate Research Fellow — *Nov. 2024 – Nov. 2025*

- Developed and validated algorithms for sleep disorder assessment from polysomnography; first-author article in *Algorithms* (paper).

Politecnico di Torino — *Turin, Italy*

Research Internship — *Mar. 2022 – Jun. 2022*

- Reviewed computational tools for molecular interaction analysis of small non-coding RNAs; presented at DETERMINED 2022 (paper).

Education

Politecnico di Torino — *Turin, Italy*

Master's Degree in Biomedical Engineering (eHealth orientation) — *Sep. 2022 – Dec. 2024*

- Main Courses: Technologies for Telemedicine, Biomedical Signal Processing, Artificial Intelligence in Medicine, Telerehabilitation
- Thesis: deep learning for vocal signal analysis in Parkinson's disease; first-author article in *Biomedical Signal Processing and Control* (paper).
- GPA: 29.33/30, Final Grade: 110/110 cum laude

Trinity College Dublin — *Dublin, Ireland*

Erasmus Exchange Program — *Sep. 2023 – Jan. 2024*

- Main Courses: Experimental and Research Methods in Biomedical Engineering, Biomaterials, Neural Signal Analysis, Deep Learning
- Results: First-class honours in all courses

Politecnico di Torino — *Turin, Italy*

Bachelor's Degree in Biomedical Engineering — *Sep. 2019 – Jul. 2022*

- Main Courses: Bioimaging, Electrical Circuits, Implantable Active Devices, Signal Analysis, Biomedical Instrumentation and Safety
- GPA: 28.29/30, Final Grade: 110/110 cum laude

Leadership & Outreach

HKN Mu Nu Chapter, IEEE — *Turin, Italy*

Corporate Training Area — *Mar. 2024 – Sep. 2026*

- Organized workshops and seminars for student professional development.

WeAreHERe - PoliWO, Politecnico di Torino — *Turin, Italy*

Student Mentor — *Sep. 2023 – Sep. 2024*

- Supported initiatives promoting female enrollment in engineering by providing mentoring and organizing awareness events.

Honors & Awards

2026	**PhD Fellowship Strategic Basic Research**, Research Foundation Flanders (FWO)	*Belgium*
2019–22	**Awardee**, Young Talent Project, Politecnico di Torino (Top 200 students, badge)	*Turin, Italy*
May 2017	**Winner**, "Una Settimana da Ricercatore" Competition, University of Salento	*Lecce, Italy*

Skills

Programming Python (PyTorch), MATLAB, C
Methods EEG analysis, Biomedical signal processing, Deep learning
Tools LaTeX, GraphPad, SolidWorks
Languages Italian (Native), English (Fluent, C1)